Day 3: The geometry of Eigen - Theory & PCA

Theory Assignment:

~~Task A:~~

Problem 1: The characterstics equation.

Task A: Formally explain why the determinant of the matrix $(A - \lambda I)$ must equals exactly zero.

Task B: If $\det(A - \lambda I) \neq 0$, what is the only possible solution of vector v? Why is this solution useless in ML?

Solution:

Task A: The eqn $(A - \lambda I)v = 0$. is a homogenous linear system. To obtain valid eigenvector, we require a non trivial solution $v \neq 0$.

geometricly $\det(A - \lambda I) = 0$ means transformation collapes space into a lower dimension. and create a non trivial null space.

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Task B: If $\det(A-\lambda I) \neq 0$, then $(A-\lambda I)$ is Invertable. In that case:

$$(A-\lambda I)v = 0 \Rightarrow v = (A-\lambda I)^{-1} \cdot 0 = 0.$$

The solution is 0 vector.

This solution is useless in ML because:

- The 0 vector have no direction.
- In PCA, Optimization, we need to meaningful directions that describe variance, curvature or structure.

Problem 2: Geometric Distortion and eigenvalues.

Task A: Imagine you have a circle of point around dataset in a 2D plan. you Apply transformation T matrix on the data set, T matrix has two eigenvector v_1, v_2. eigenvalue $\lambda_1 = 4$ & $\lambda_2 = 0$.

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Describe exactly what happens to the geometric shape of the dataset after transformation T is applied.

Task B: matrix T is functionally performing dimensionality reduction. explain how the eginevalue $\lambda_2 = 0$ dictates this collaps of space.

Task C: What is the det of matrix T? Formally justify your Answer based on eigenvalues.

Solution:

Task A: The orginal Dataset is a circle centered at the origin. under a liner transformation:

- Along eignvector v_1 with eigenvalue $\lambda_1 = 4$, all component are stretched by a factor of 4.
- eigenvector v_2 with eigenvalue $\lambda_2 = 0$, all component are collapsed to zero.

The transformation projects the 2D circle dataset

onto a 1D subspace along with the axis.

Task B: An eigenvector of 0 means all vector ~~using~~ along that specific eigenvector span are mapped to the origin. The transformation destroy the dimension. reducing Data from 2D to 1D.

Task C: $\det(T) = 0$.

The det is the product of the ~~eigenvectors~~ eigenvalues $(\lambda_1 \times \lambda_2) = 0$.

2D area collapses to a 1D line.

problem 3: The Spectral Theorem and PCA.

Task A: State the specific geometric relationship between eigenvectors of symmetric matrix.

Task B: In PCA, the eigenvectors of the covariance matrix become our new features axes. Based on your answer to Task A. explain why this property is highly desirable for ML feature engineering?

Solution:

Task A: The eigenvectors of a symmetric matrix are strictly orthogonal. (perfectly perpendicular to other)

Task B: orthogonal axes represent zero linear correlation. By projecting our data onto these orthogonal eigen vectors, we guarantee that our new features are uncorrelated with each other.

This strips out all redundancy in the dataset, making it incredibly efficient for ML models.

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